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Literature Review

مراجعة الأدبيات

GlobeClass — AI-Powered Satellite Image Land Cover Classification

1. Background & Context

Remote sensing and satellite imagery have become indispensable tools for monitoring Earth's surface. With the increasing availability of satellite data from missions such as Sentinel-2 (European Space Agency), the need for automated land cover classification systems has grown significantly. Manual interpretation of satellite images is time-consuming, subjective, and impractical at scale — making deep learning-based automation a critical advancement.

Land cover classification involves categorizing each region of a satellite image into predefined classes such as forests, water bodies, urban areas, agricultural land, and more. Accurate land cover mapping supports a wide range of applications including:

- Urban planning and development monitoring
- Agricultural productivity assessment
- Environmental and climate change monitoring
- Disaster response and damage assessment
- Water resource management
- Biodiversity conservation

The advent of deep learning, particularly Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), has revolutionized the accuracy and efficiency of satellite image classification, achieving human-level or beyond-human-level performance on standard benchmarks.

2. Previous Work & Related Studies

2.1 Traditional Machine Learning Approaches

Early approaches to land cover classification relied on handcrafted features and traditional machine learning algorithms:

Support Vector Machines (SVMs): Used spectral and textural features extracted from satellite imagery for pixel-level classification. Mountrakis et al. (2011) provided a comprehensive review of SVM applications in remote sensing, noting accuracy levels of 70-85% depending on feature engineering quality.

Random Forests: Belgiu and Dragut (2016) surveyed Random Forest classifiers in remote sensing, demonstrating their robustness for multi-class land use classification but highlighting limitations in capturing spatial hierarchies.

K-Nearest Neighbors (KNN): Simple distance-based classifiers were used in early land cover studies, but were limited by the curse of dimensionality in high-resolution imagery.

Limitations: These methods require extensive manual feature engineering, struggle with complex spatial patterns, and do not scale well to large datasets.

2.2 Convolutional Neural Networks (CNNs)

CNNs revolutionized image classification by automatically learning hierarchical spatial features:

He et al. (2016) — Deep Residual Learning (ResNet): Introduced skip connections to train very deep networks (up to 152 layers), solving the vanishing gradient problem. ResNet50, with its 25.6 million parameters, became a standard backbone for transfer learning across visual recognition tasks.

Helber et al. (2019) — EuroSAT Dataset: Created the EuroSAT dataset specifically for land cover classification using Sentinel-2 satellite images. They demonstrated that CNNs (including ResNet50) could achieve up to 98.57% accuracy on this benchmark using transfer learning from ImageNet.

Transfer Learning: Yosinski et al. (2014) showed that features learned from large-scale natural image datasets (like ImageNet) transfer effectively to domain-specific tasks, significantly reducing training data requirements and improving convergence speed.

2.3 Vision Transformers

Transformers, originally designed for natural language processing, were adapted for computer vision:

Dosovitskiy et al. (2020) — Vision Transformer (ViT): Proposed treating images as sequences of patches and applying the standard Transformer architecture. ViTs achieved competitive results with CNNs on ImageNet classification.

Liu et al. (2021) — Swin Transformer: Introduced a hierarchical vision transformer that computes self-attention within shifted local windows. Key innovations include:

- Linear computational complexity relative to image size (vs. quadratic for standard ViTs)
- Multi-scale feature representation through hierarchical patch merging
- Swin-Base achieved 83.5% top-1 accuracy on ImageNet with 88M parameters
- Particularly well-suited for dense prediction tasks common in remote sensing

2.4 Foundation Models for Earth Observation

A recent paradigm shift involves pre-training large models on massive Earth observation datasets:

Jakubik et al. (2023) — Prithvi-EO: IBM and NASA developed the Prithvi foundation model, a Vision Transformer pre-trained on NASA's Harmonized Landsat and Sentinel-2 (HLS) multi-spectral satellite imagery using Masked Autoencoder (MAE) self-supervised learning. Key characteristics:

- Trained on globally distributed satellite data across multiple spectral bands (B, G, R, NIR, SWIR1, SWIR2)
- Designed for versatile downstream tasks: classification, segmentation, change detection
- Demonstrates strong zero-shot and few-shot generalization to unseen geographic regions
- The Prithvi-EO 2.0 model with 600M parameters represents a significant advancement in geospatial AI

2.5 Web-Based Remote Sensing Applications

Several existing systems provide web-based satellite image analysis:

Google Earth Engine: A powerful cloud platform for planetary-scale geospatial analysis, but requires specialized coding knowledge and is not focused on real-time interactive classification.

Microsoft Planetary Computer: Provides access to global environmental datasets with built-in analysis tools, targeting researchers and data scientists rather than general users.

EO-Learn (Sentinel Hub): A Python library for processing satellite imagery with machine learning pipelines, but lacks a consumer-facing web interface.

3. Technologies Reviewed

Technology	Type	Relevance to Project
ResNet50	CNN Architecture	Backbone for transfer learning — proven on EuroSAT with 98%+ accuracy
Swin Transformer	Vision Transformer	Hierarchical attention mechanism ideal for multi-scale remote sensing features
Prithvi-EO 2.0	Foundation Model	Pre-trained on global satellite data, designed for Earth observation tasks
PyTorch	Deep Learning Framework	Industry-standard framework for model training and inference
timm	Model Library	Provides pre-trained weights and architecture implementations for 700+ models
terratorch	Geospatial ML Library	IBM's library for registering and using Prithvi models within the timm ecosystem
Flask	Web Framework	Lightweight Python web framework for building REST APIs
Three.js	3D Graphics Library	WebGL-based library for rendering the interactive 3D Earth globe
Leaflet.js	Mapping Library	Open-source library for interactive satellite maps with ESRI tile integration
EuroSAT	Benchmark Dataset	Standard dataset for satellite image land cover classification (27K images, 10 classes)
ESRI World Imagery	Satellite Tile Service	Provides real-time satellite imagery for any location globally
Sentinel-2	Satellite Mission	Source of the EuroSAT dataset — ESA's multi-spectral imaging satellite

4. Gaps in Existing Work

Despite significant progress in satellite image classification, several gaps remain that GlobeClass addresses:

#	Gap	How GlobeClass Addresses It

1	Lack of user-friendly interfaces: Most existing tools (Google Earth Engine, EO-Learn) require programming expertise and are designed for researchers, not general users.	GlobeClass provides an intuitive, visually stunning web interface with a 3D globe, drag-and-drop image upload, and one-click classification accessible to anyone.
2	Single-model systems: Most deployed classification tools use a single model architecture without offering users a choice.	GlobeClass deploys three distinct architectures (ResNet50, Swin Transformer, Prithvi-EO) and allows users to switch between them in real-time to compare results.
3	No real-time satellite capture integration: Existing classification systems typically require users to pre-download satellite images from separate platforms.	GlobeClass integrates ESRI World Imagery to capture live satellite views from any coordinates, directly feeding them into the classification pipeline.
4	Limited accessibility and language support: Most remote sensing tools are English-only and lack Progressive Web App capabilities.	GlobeClass features full English/Arabic bilingual support and PWA functionality for offline access and mobile installation.
5	No foundation model deployment for classification: While Prithvi-EO was released as a research model, practical deployments in web-based classification interfaces are scarce.	GlobeClass adapts and deploys the Prithvi-EO 2.0 foundation model with a custom RGB-to-6-band channel adapter, making it accessible through a web API.
6	Opaque classification results: Many systems output only the top predicted class without detailed confidence breakdowns.	GlobeClass provides full 10-class probability distributions with interactive charts, animated breakdowns, and classification logging history.

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